

What Does Expected Goals (xG) Actually Tell Us?

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Internationally, football also known as soccer is one of the most followed sports in the world. The highest level football leagues, and the most talked about leagues, in the world are the Men's Big 5 European Leagues which include the Premier League, La Liga, Serie A, Bundesliga, and Ligue 1 (Statista, 2024). The quality of play is high, and we chose to look at one of the most discussed statistics in football expected goals also known as xG.

xG is the expected goals including penalty kicks (not shootouts unless stated), or in other words, the probability of a shot resulting in a goal. Opta Analyst created xG and introduced it into the top leagues in 2012 (Opta Analyst, 2023). There are 20 variables in the Opta model for their xG calculation relating to field position of the player, body part used, and other variables that are mainly based on the circumstances on the pitch based off thousands of pieces of past data (Opta Analyst, 2023). While xG is widely used, it is also often a misnomer. xG means expected goals, but it is specifically analyzing the likelihood of scoring based off the quality of the opportunity. xG does not mean the expected goal count or how many goals a team should've scored during a match (Opta Analyst, 2023). xG can be applied to analyze if a team produces a lot of chances or if they are just clinical finishers. An xG of 0 would be a certain miss, and an xG of 1 would be a certain goal. When looking at the xG from a game, sometimes there are large discrepancies between xG and the final score. The

goal of this project is to see what xG is good at predicting and modeling if it can't be a predictor for goals. The importance of this research is to help fans, pundits and teams understand what xG is explaining about a team's offensive qualities. Additionally, to understand how xG can be used as a statistical tool to understand clinicalness, luck and efficiency.

Data Collection

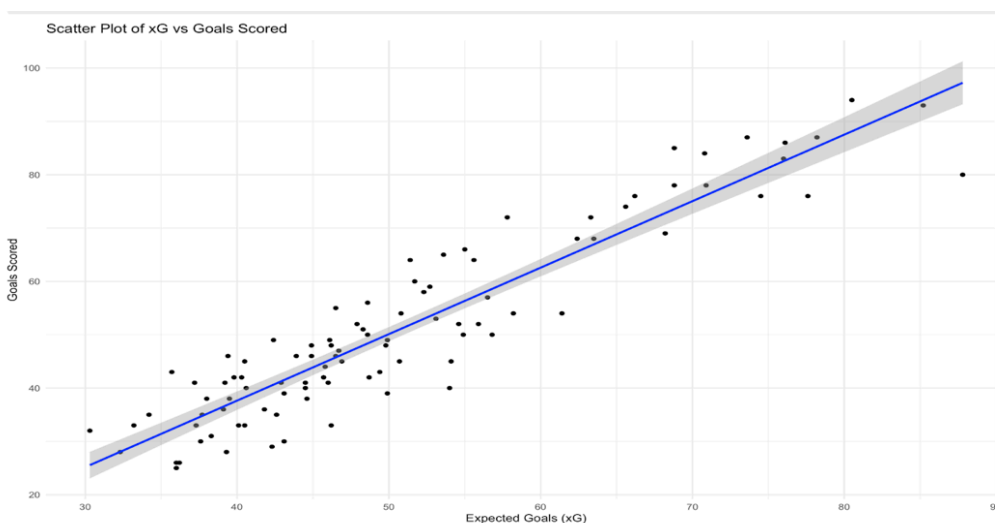
The dataset was obtained from fbref.com for the Men's Big 5 European Leagues 2023-2024 season. The data used was the squad stats standard stats and the squad stats shooting stats from the 2023-2024 season. Originally we used worldfootballR to get the datasets off of fbref.com but there were some issues with worldfootballR. In worldfootballR there are 192 teams listed separately as team and opponent, when in reality there are 96 teams which made the data hard to scrape. The data also did not include the end of season league finish which was a key response variable later in the research project, so because of these issues, we decided not to use worldfootballR. Instead, we created an Excel spreadsheet with the data pasted from their website. We used worldfootballR to corroborate with the Excel spreadsheet to make sure there was no missing data by randomly selecting 5 observations from the excel spreadsheet and comparing it to the worldfootballR dataset. After that the spreadsheet was converted into a csv. file where we corroborated the data again. After the data was read into R, we used the R package ggplot2 to help with graphic visualizations.

xG as a Predictor

We started our research by looking at the relationship between xG and goals to see if there was some basis to the common misinterpretation of xG. The scatterplot, in Figure 1, showed that there was a positive, linear relationship between xG and goals scored that is significant. We also ran a model summary, and the correlation was 0.928 showing a strong positive relationship. The multiple R^2 value of 0.8614 means that xG as a predictor explains 86% of the variability within the regression and the high adjusted R^2 value of 0.8599 means that even after considering the number of predictors within the model xG still explains nearly 86% of the variability in the regression. Lastly, a small p-value of 2×10^{-16} means that xG is a significant predictor. This means that there is some validity in people believing in this misnomer, but xG is not intended as a predictive statistic and is generated after each game therefore it can't be. Additionally, goals scored is a variable in creating xG which leads to circular logic if xG is treated as a predictor. While xG may not predict goals, it can still be useful for assessing a team's overall performance and potential over a longer period. Thus, leading us to explore how xG could influence league finish.

Figure 1

Scatter Plot of xG vs. Goals Scored

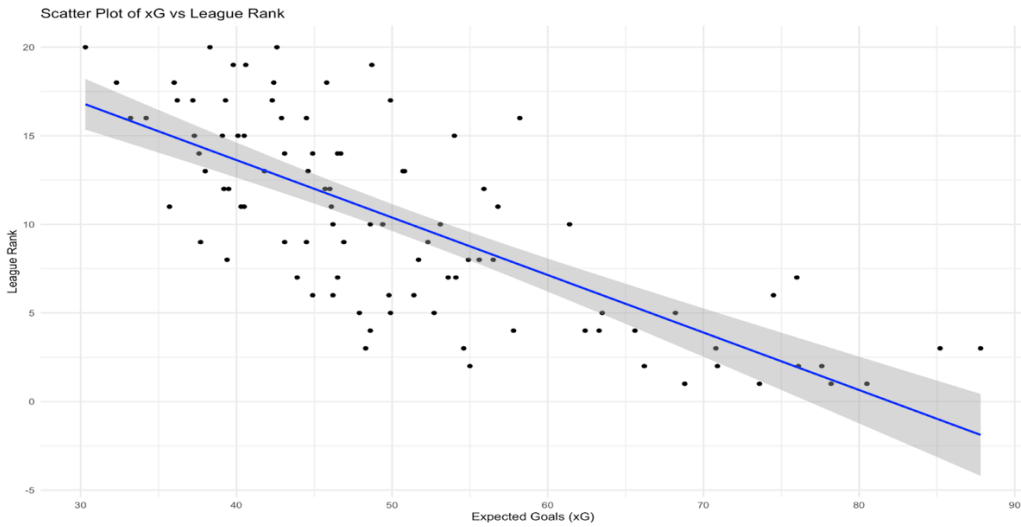


Note. This graph shows the regression line for a model with expected goals (xG) as a predictor variable for goals scored for each of the Men's Big 5 Leagues teams in the 2023-2024 season.

We plotted xG versus the league finish of each team in the Big 5 Leagues in each of their respective leagues. This scatter plot, in Figure 2, has a negative, linear relationship, which could be expected since a higher xG could signify better league ranking, and we are assuming a team with more scoring opportunities would finish better. The correlation has a value of -0.748 which is a strong correlation. The multiple R^2 value was 0.56 , and the adjusted R^2 value was 0.5553 so xG explains about 55.5% of the variability in league rank in the regression. The p-value is 2×10^{-16} which signifies a statistically significant relationship between xG and league rank. This relationship shows that with a higher xG, a team would finish at a lower value place in league standings, which also means a higher, or better, place finish in the league.

Figure 2

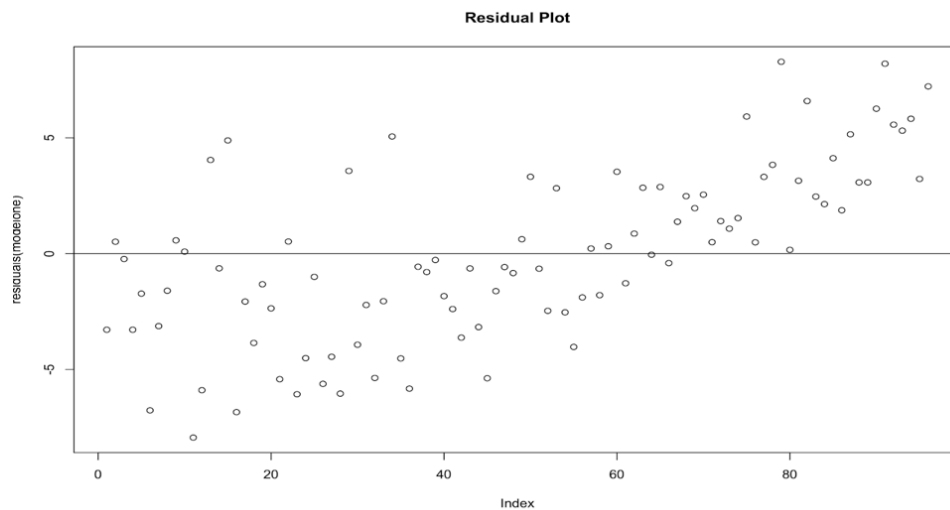
Scatter Plot of xG vs. League Rank



Note. This graph shows the regression line for a model with expected goals (xG) as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Figure 3

Residual Plot of xG Model Predicting League Rank



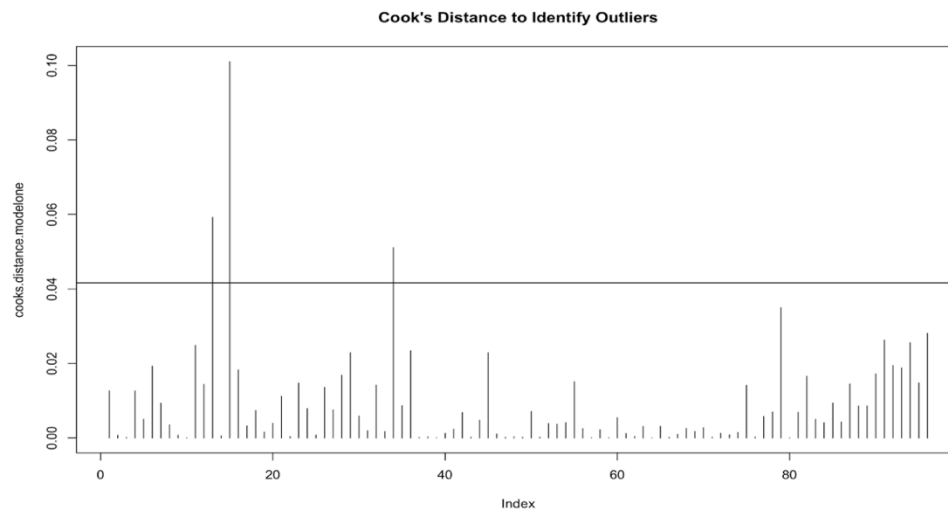
Note. This figure shows the residual plot for the model with expected goals (xG) as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

We decided to create a residual plot next to see if xG as a predictor could explain enough to cause homoscedasticity. However, after creating the residual plot seen in Figure 3, there is a clear violation of homoscedasticity since there is a linear relationship meaning that xG does not fully explain the league outcome. After realizing this we decided to see if influential points had any influence on xG as a predictor.

We used Cook's Distance to identify influential points for the model of xG versus league rank since there was a linear relationship in the residual plot in Figure 3. Above the line, there are shown to be three influential points in Figure 4. These three points were Bayern Munich, Liverpool, and Newcastle United. We then removed the influential points and then ran the model again without them. We looked at the DFBETAS for the

Figure 4

Cook's Distance for Outlier Detection of xG Model Predicting League Rank

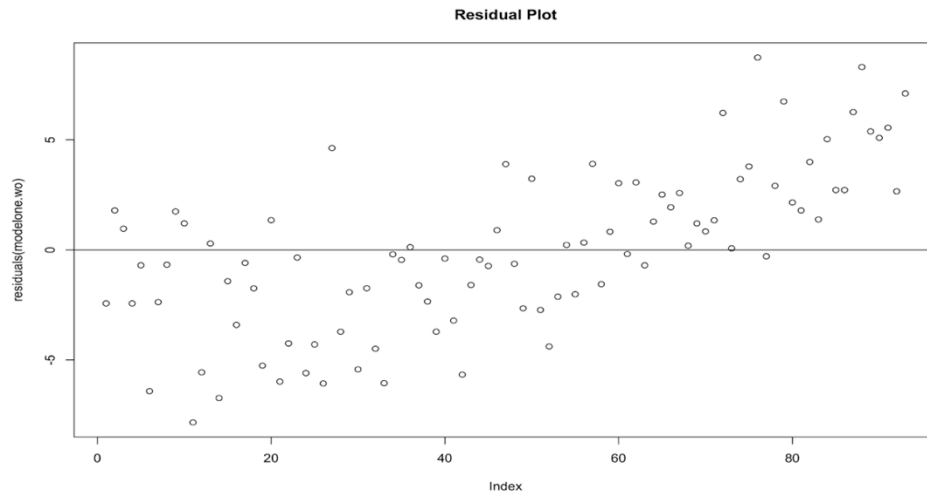


Note. This figure shows the Cook's Distance for outlier detection for the model with expected goals (xG) as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

influential points of the first model, and after their removal, each intercept changed by a number greater than one meaning that the influential points had an impact on the original model. The new model without influential points had an adjusted R^2 value of 0.5673, a multiple R^2 value of 0.572, which means xG explains around 57% of league rank in the regression without the influential points in the model. The p-value was 2×10^{-16} which again means that there is a significant relationship between xG and league rank. The removal of influential points improved our model and made xG a better predictor of league finish therefore indicating a better fitted model. However, when creating another residual plot without the influential points, there is still heteroscedasticity within the model, as seen in Figure 5, meaning xG does not explain everything about league finish when in a regression.

Figure 5

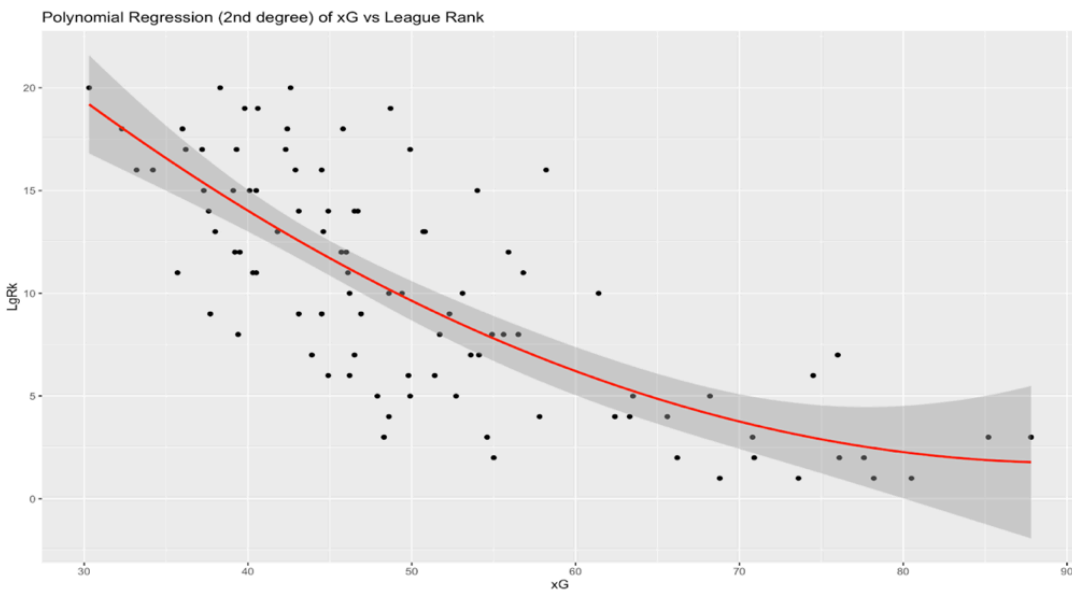
Residual Plot of xG Model Predicting League Rank After Outliers Removed



Note. This figure shows the residual plot for the model with expected goals (xG) as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season after outliers were removed.

Figure 6

Scatter Plot of (2nd Degree) xG Model Predicting League Rank



Note. This graph shows the regression line for a model with expected goals (xG) to the second degree as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

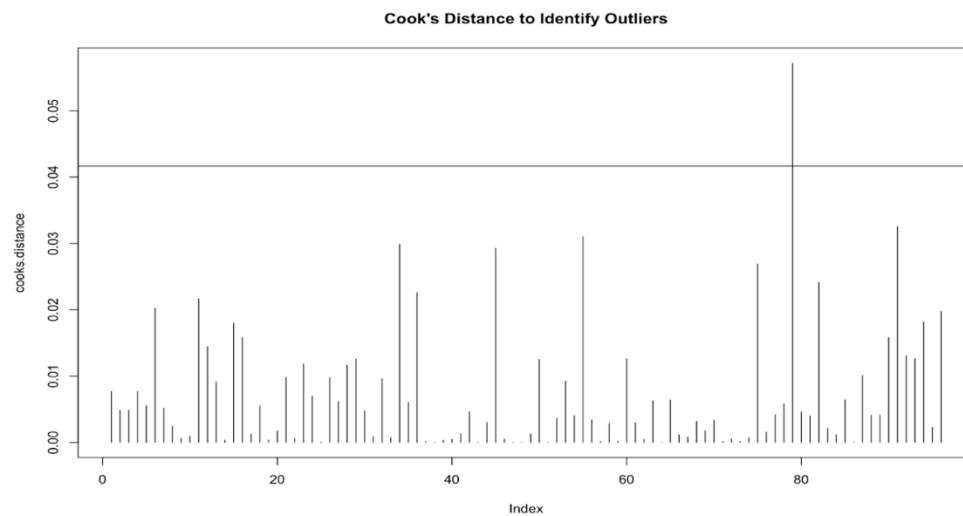
Looking at the scatter plot in Figure 2 closely, there is a potential slight exponential relationship, therefore, we considered higher power model fitting to improve the regression. The new model can be seen in Figure 6. The relationship R^2 values increased when we squared xG to an adjusted R^2 value of 0.5783 and a multiple R^2 value of 0.5872, which explains around 58% of the variability in the regression. xG squared was a significant predictor with a p-value of 2×10^{-16} . This all signifies that xG has a better exponential relationship with league finish than one that is linear. We also decided to use Cook's Distance on this model, and found one influential point which was Brentford, as seen in Figure 7. After removing the influential point, the new adjusted R^2 value was 0.6063, and the multiple R^2 value was 0.612, which were increases from the previous model, and p-value of 2×10^{-16} . We then used DFBETAS on this model, and did not find that removing the influential point had a meaningful impact on the model. The intercepts did not meaningfully change leading us to keep the influential point in the dataset.

We also decided to try cubing xG to see if the fit would be better for league rank since squaring the model previously improved it, and the scatter plot can be seen in Figure 8. The adjusted R^2 value for the third power model was 0.5752, the multiple R^2 value was 0.5886. These R^2 values indicate that xG^3 explains around 58% of variability of league rank in the regression. xG^3 was a significant predictor of league rank with a

p-value from the regression of 2×10^{-16} . These R^2 values decreased from the squared xG model, so this means that the cubed xG model explains less of the variability in league rank than the squared model. We also tried using Cook's Distance on this model to see if there was any possibility of improvement. There was one influential point that was

Figure 7

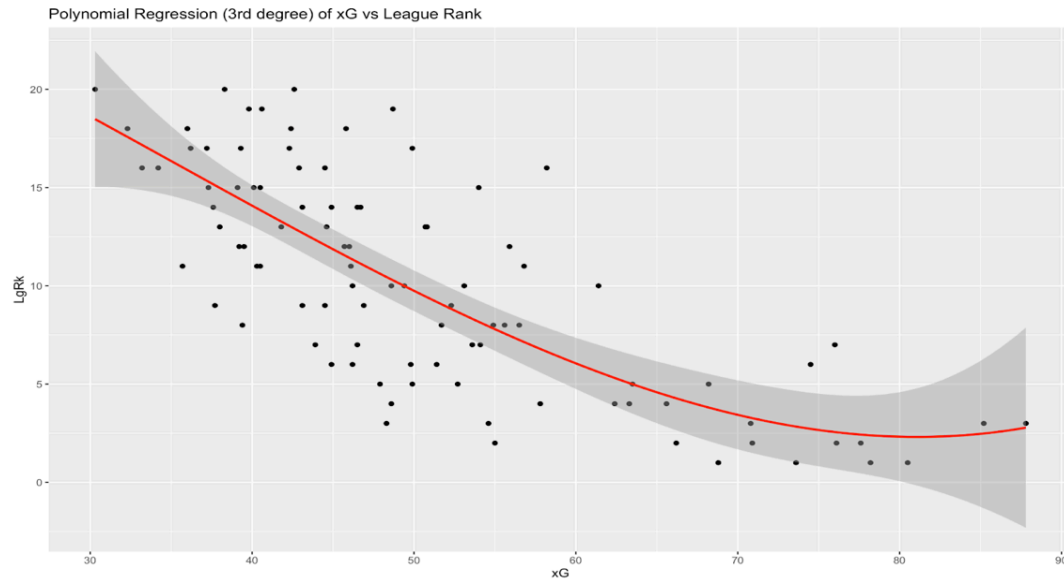
Cook's Distance for Outlier Detection of xG^2 Model Predicting League Rank



Note. This figure shows the Cook's Distance for outlier detection for the model with xG^2 as a predictor variable for league rank for each of the Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Figure 8

Scatter Plot of (3rd Degree) xG Model Predicting League Rank

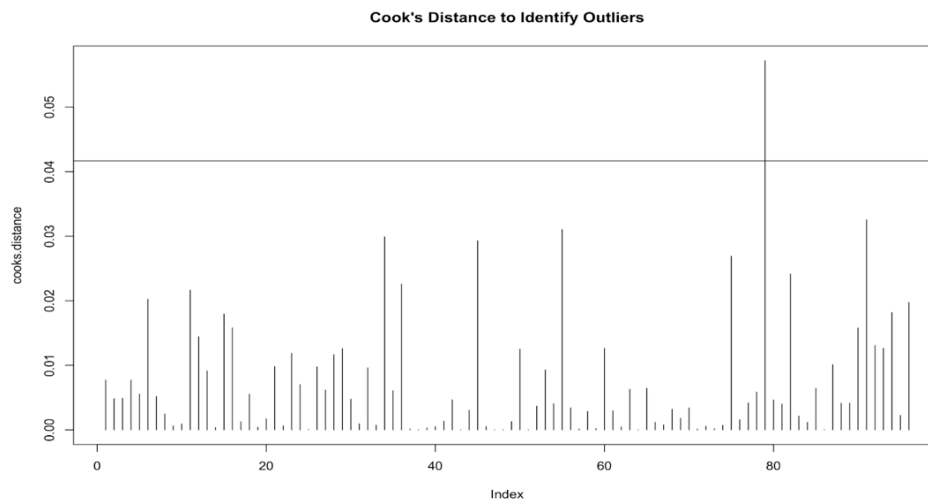


Note. This graph shows the regression line for a model with expected goals (xG) to the third degree as a predictor variable for league rank for each of the Men's Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Brentford again, as observed in Figure 9. After removing this influential point from the cubed model, the new adjusted R^2 value was 0.6015, and p-value was significant for xG^3 as a predictor for league rank at 2.2×10^{-16} . The multiple R^2 value was 0.6142. These R^2 values mean that xG^3 explains about 60.7% of league rank. These adjusted and multiple R^2 values were an improvement from the original cubed model, but not an improvement from the squared model with the influential point removed. Additionally, the DFBETAS showed that there was no meaningful change with the influential point removed. The model did not improve fit after being cubed, so we decided to stop testing more higher power models.

Figure 9

Cook's Distance for Outlier Detection of xG^3 Model Predicting League Rank



Note. This figure shows the Cook's Distance for outlier detection for the model with xG^3 as a predictor variable for league rank for each of the Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Luck Factor in League Outcome

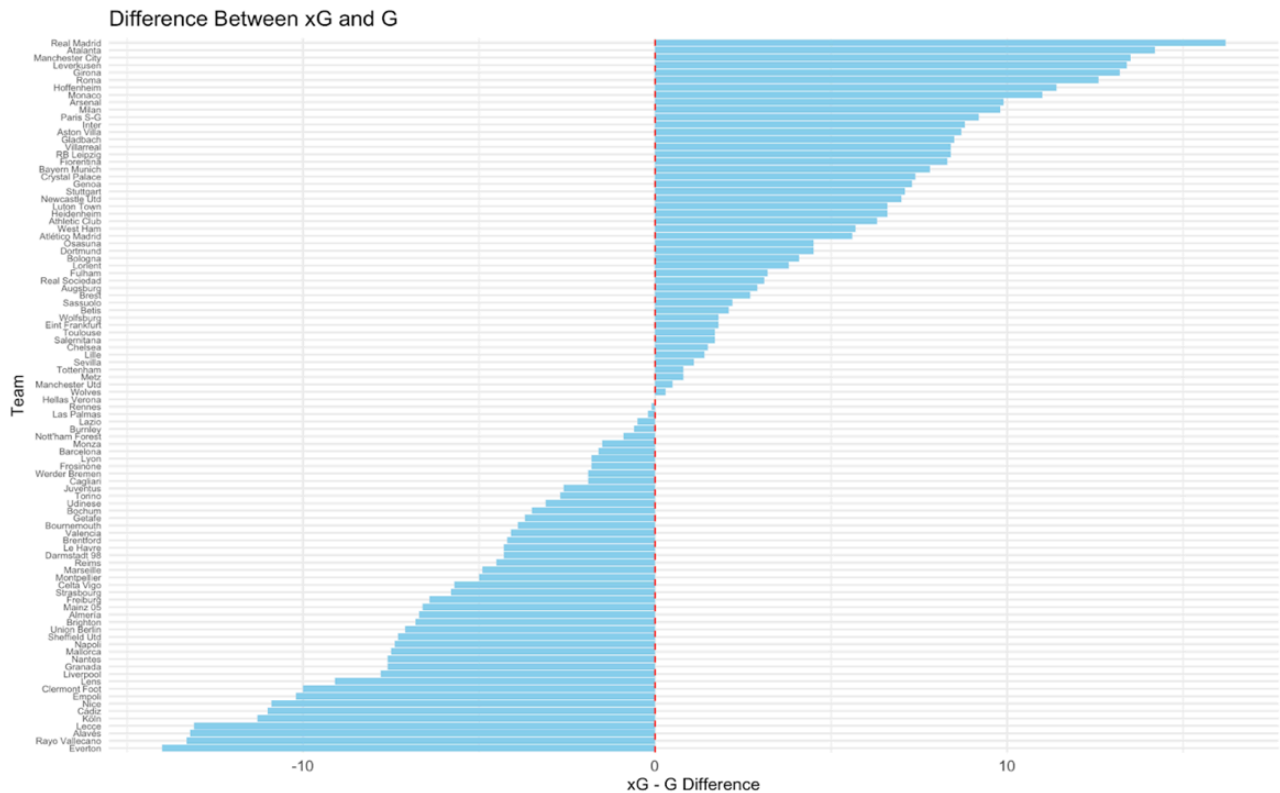
xG is a measurement of the quality of chances, but a good chance is not always converted, and a bad chance can be converted. Since xG has a moderate correlation with league finish we wanted to use xG as a variable for luck and see if luckier teams finished higher in the league. By using the difference between goals scored and xG , we considered the remaining value to be luck. This metric assumes that goals scored above or below the xG are influenced by factors that xG does not account for, such as goalkeeper errors, defensive mistakes, or exceptional finishing ability. In this context, 'luck' captures the variability in match outcomes that cannot be predicted purely by the quality of chances created. A positive value of this difference would mean a team is

luckier because they scored more goals than expected based on the opportunity, and a negative value of the difference would signify that a team is unlucky because they are not scoring enough goals compared to their scoring opportunities. We generated the differences of goals and xG for each of the Big 5 Leagues and created a visual to represent the differences for each team to be observed in Figure 10.

Based on these differences, the luckiest team out of all of the Big 5 Leagues was Real Madrid, and the unluckiest team was Everton. Real Madrid finished first in La Liga, Everton, the unluckiest team, finished last in the Premier League and ended up being relegated. Manchester City was the third-luckiest team out of the Big 5 Leagues and luckiest Premier League team, and they finished first in the Premier League. Bayern Leverkusen who was the fourth luckiest team in the Big 5 Leagues and was the luckiest Bundesliga team also finished first in their league. If isolating these four results, it would appear luck is significant at predicting league finish. On the other hand, not all

Figure 10

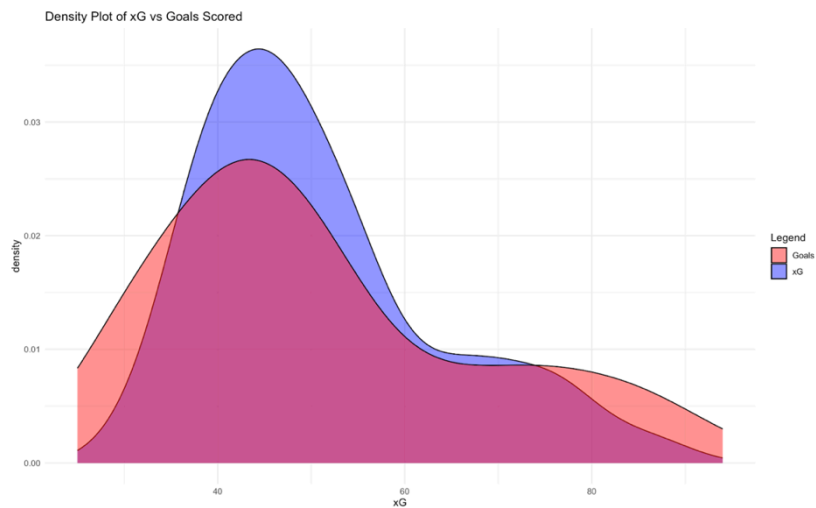
Differences Between Goals Scored and xG



Note. This figure shows the differences between goals scored and expected goals (xG) for each of the Big 5 Leagues teams for the 2023-2024 season.

teams' league finish was greatly influenced by their luck, like Barcelona. FC Barcelona finished in second but was a slightly unlucky team. Similarly, Liverpool, finishing third in the Premier League, was very close to the bottom of the list in terms of luck, being the 11th unluckiest team out of 96 teams.

We created a density plot to represent the xG and goals scored, and how the differences alter their values. The density plot in Figure 11 shows that goals scored, in red, is much less than xG, in blue. This generally makes sense because goals may not

Figure 11*Density Plot of Goals Scored and xG*

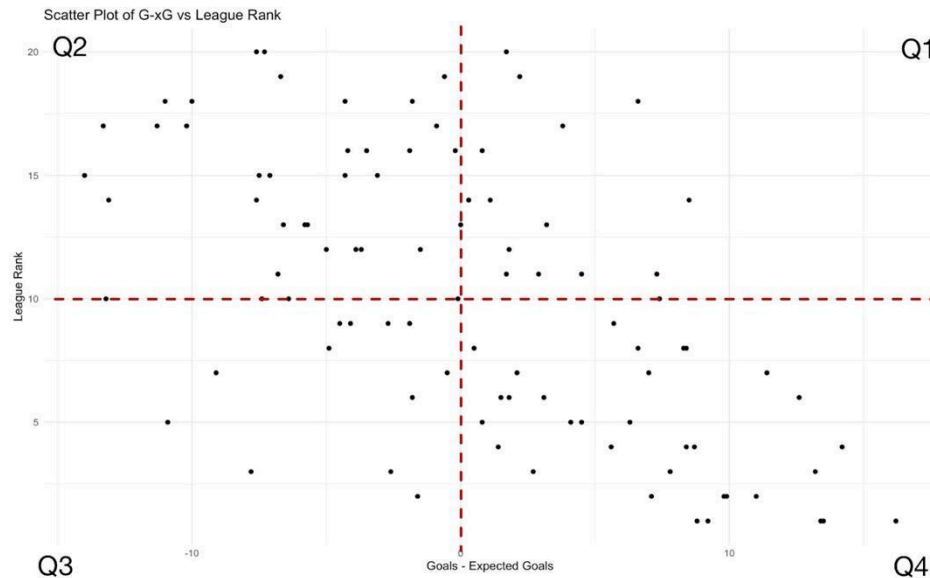
Note. This figure shows a density plot of goals scored (in red) and expected goals (xG in blue) for each of the Big 5 Leagues teams for the 2023-2024 season.

be scored with every scoring opportunity, and there will naturally be discrepancies between the two values, but with discrepancy being so large we wanted to see if any of the discrepancy can be explained by luck.

With the idea of luck being a factor for outcomes, we decided to investigate if luck could explain league finish. We created a scatter plot for league rank versus luck. When looking at the scatter plot, we divided it into four quadrants starting with the top right being Quadrant 1, then moving counterclockwise with Quadrant 2 being the top left, Quadrant 3 being the bottom left, and Quadrant 4 being the bottom right. For the scatter plot of luck versus league rank, Figure 12, a lot of the data falls into Quadrants 2 and 4, and there appears to be somewhat of a linear, negative relationship between luck and league rank.

Figure 12

Scatter Plot of Goals Scored - xG vs. League Rank



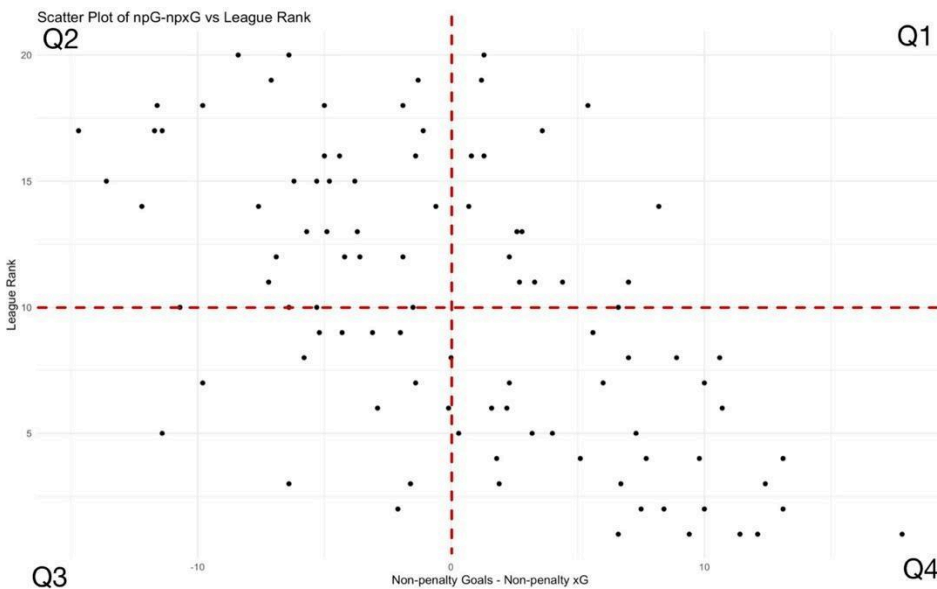
Note. This figure shows a scatter plot of goals scored-xG (luck factor) versus league rank for each of the Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Since penalty shots can be a major goal-scoring opportunity, we decided to investigate luck while excluding penalty shots/goals. We used the difference between non-penalty (NP) goals and NPxG to create a non-penalty luck factor and created a scatter plot of NP luck versus league rank. Following the same division into quadrants for Figure 13 as the previous scatter plot, many, if not most, of the data points fell into Quadrants 2 and 4. Quadrants 1 and 3 are relatively low on data points very prominently. With the majority of the data points falling in Quadrants 2 and 4, we considered that this signified a linear, negative relationship between luck and league finish. Because of this evidence, we concluded that luck can explain league finish. The

luckiest teams, with a luck above 10, all finished towards the top of their league (in the top 10). We generated a similar display of difference between NP goals and NPxG for

Figure 13

Scatter Plot of Non-Penalty Goals Scored – Non-Penalty xG vs. League Rank



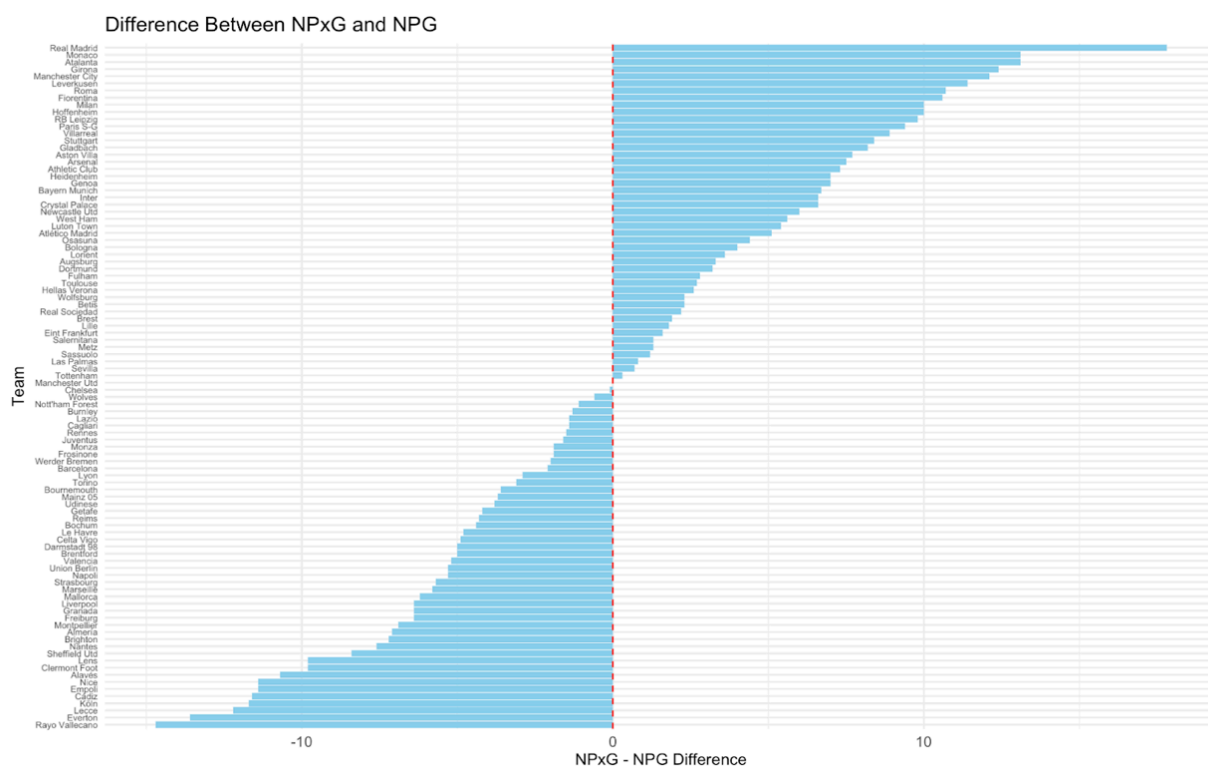
Note. This figure shows a scatter plot of NP goals scored - NPxG (non-penalty luck factor) versus league rank for each of the Big 5 Leagues teams in their respective leagues for the 2023-2024 season.

Figure 14. As before, the results are mainly similar with Real Madrid being the luckiest at the top of this list, but by an even greater amount, and Rayo Vallecano being the unluckiest team and Everton above them by one position. This could signify that penalty goals are a factor in what a team's luck. Penalty kicks can be considered a gray area for adding to the luck of a team since they tend to be a good scoring opportunity, but not a given goal necessarily. The xG value given for a penalty shot is 0.79 which is a high conversion rate (FBref, (n.d.)). Similarly, Monaco is the second-luckiest team in the luck rankings of NP goals/xG but was the eighth-luckiest team in the overall goals

and xG difference rank. Clearly penalty shots had enough influence on the data for the luck rankings we generated to have shifted around somewhat, which is why we felt it important to include both data sets and graphs to analyze.

Figure 14

Differences Between Non-Penalty Goals Scored and Non-Penalty xG



Note. This figure shows the differences between non-penalty goals scored and non-penalty expected goals (xG) for each of the Big 5 Leagues teams for the 2023-2024 season.

Discussion & Conclusion

We compared the xG from the teams in the men's 2023-2024 Big 5 European Leagues. After finding the outcome of final table position/league finish of the 2023-2024 season, we investigated how this compares to their xG.

We have found that while xG is not technically able to be a predictor variable for goals scored since it is a post-game statistic, it can still be used to predict overall league finish and there is a relationship between the two. We also can use xG to determine how much of teams' success based on their league finish is due to luck. Luckier teams are shown to finish at the top of the pack. xG is not an exhaustive predictor and neither is luck player quality is not evenly dispersed across leagues.

There was movement in our luck rankings when comparing their luck of all goals minus xG versus the luck of only non-penalty goals minus non-penalty xG. Because of the difference in luck with the penalty goals included and not included, we think it is important to address this difference in future research. Specifically analyzing how much penalty shots influence the scoring and luck of a team. How does being awarded (more) penalties influence a team's play and/or scoring? This could influence teams to change their fouling behavior in the box.

Another thing that we noticed was that two of the teams that were influential points in the Cook's Distance of the luck versus league rank model were from the Premier League, and all the teams that were influential finished in the top 10.

We could further test this analysis on other seasons from the Big 5 or on other leagues. Season to season with the same leagues could show similarities and would potentially reinforce the results. Additionally, we could create a time series and see if time influences xG to see if teams today are better at creating quality chances, then in the past. Applying the analysis to other leagues would be interesting to see the

outcome, especially if it is a league with less skill objectively. How different would the luck factor be in a league with “less skill” to explain points or rankings, or would it be similar if there are similar levels of competition between teams? Another idea was to see if we could build a predictive model based on skill or on-field data, possibly including xG, but then also testing other variables that come from off the field, such as salary, or even looking at home versus away having an impact on points for the season. Because of knowing that xG could be used as a predictor, we could use these other variables in a multiple regression to see if lucky teams are richer, or other combinations of these variables that could factor into league outcomes. We could also use this to look at xG as a variable for wins.

One limitation of this study would be that we did not look at how the Big 5 Leagues teams compare or play each other, which we could do by using data from the Champions League. The Champions League data would be different from the regular season, and we could have more understanding of how teams compare from league to league. We could use data from the Champions League to see how luck spans across the teams, and to see if teams’ luck rankings change from their regular league season to the Champions League data. Another limitation of this study is that we only used the Big 5 Leagues, but there are many other leagues in the world. We looked at takeaways from these Big 5 Leagues teams, but there aren’t takeaways that are truly applicable to other leagues. We demonstrated something that teams could consider from other leagues, but not any direct takeaways for non-Big 5 Leagues teams. A third limitation

of this study would be that we only used one Big 5 Leagues season. We did not use multiple seasons of data, which would allow us to have a larger sample size and potentially compare the included teams with more changes in time. There could be major differences in luck for a team from season to season, or there could be some teams that seem to be having a lucky or unlucky streak and hold their place towards the top or bottom of the rankings.

References

- FBref. (2024). 2023-2024 Big 5 European leagues stats. Retrieved October 26, 2024, from <https://fbref.com/en/comps/Big5/2023-2024/2023-2024-Big-5-European-Leagues-Stats>
- FBref. (2024). 2023-2024 Big 5 European leagues shooting stats. Retrieved October 26, 2024, from <https://fbref.com/en/comps/Big5/2023-2024/shooting/squads/2023-2024-Big-5-European-Leagues-Stats>
- FBref. (n.d.). Expected goals (xG) model explained. Retrieved October 11, 2024, from <https://fbref.com/en/expected-goals-model-explained/>
- H. Wickham. ggplot2: Elegant Graphics for Data Analysis.

Springer-Verlag New York, 2016.

Opta Analyst. (2023, August). What is expected goals (xG)? Retrieved October 26, 2024, from <https://theanalyst.com/2023/08/what-is-expected-goals-xg>

RStudio Team. (2024). *RStudio: Integrated development environment for R* (Version 2024.09.0+375) [Computer software]. RStudio, PBC. <https://posit.co>

R Core Team. (2024). *R: A language and environment for statistical computing* (Version 4.4.2) [Computer software]. R Foundation for Statistical Computing. <https://www.r-project.org/>

Statista Research Department. (2024, August 2). *Big Five - statistics & facts*. Statista.com. <https://www.statista.com/topics/5909/-big-five/>

Zivkovic J (2024). `_worldfootballR`: Extract and Clean World

Football (Soccer) Data_. R package version 0.6.6.0001,

<<https://github.com/JaseZiv/worldfootballR>>.